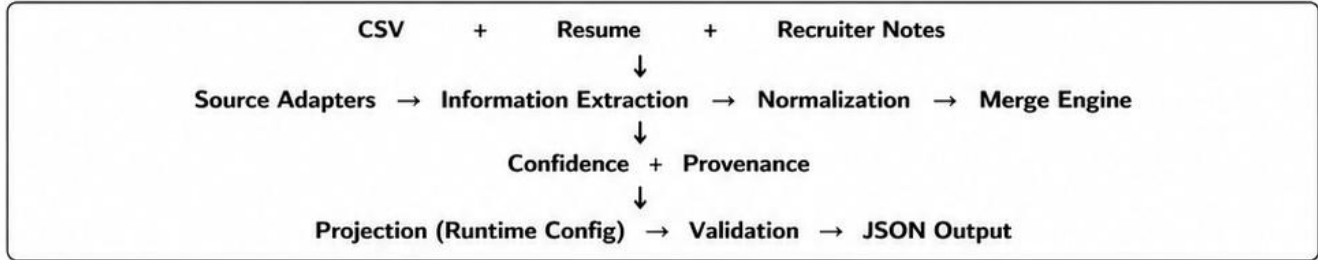


Candidate Data Transformation Pipeline

Eightfold Engineering Internship Assignment – Technical Design

Design Philosophy: Build a *deterministic and explainable* transformation pipeline that prioritizes correctness, traceability, and configurable outputs over heuristic or probabilistic extraction.

1. Pipeline Overview



2. Canonical Schema & Normalization

Canonical Record (Internal Model)	
• candidate_id	• years_experience
• full_name	• skills[]
• emails[]	• experience[]
• phones[]	• education[]
• location	• salary_expectation
• links	• availability
• headline	• provenance[]
	• overall_confidence

Normalization Rules
• Emails → lowercase
• Phones → E.164
• Skills → canonical dictionary
• Dates → YYYY-MM
• Country → ISO 3166 Alpha-2

3. Merge / Conflict Resolution Policy

Field	Strategy
Full Name	Longest complete value
Emails	Normalize + Union
Phones	Normalize + Union (E.164)
Skills	Canonicalize + Union
Experience	Concatenate + Sort
Education	Concatenate + Deduplicate
Salary Expectation	Prefer Recruiter Notes
Availability	Prefer Recruiter Notes
Location	First non-empty value
Headline	First non-empty value

Candidate Matching

- Primary key: Normalized email
- Fallback: Normalized full name

Confidence Score

Computed deterministically using source reliability, successful normalization, validation status, and cross-source agreement.

4. Runtime Projection (Configurable Output)

The pipeline always produces a fixed internal canonical record. A runtime JSON configuration defines:

- Fields to include
- Field renaming / mapping
- Per-field normalization
- Confidence / provenance inclusion
- Missing-value policy (null / omit / error)

This cleanly separates the canonical model from the output schema without code changes.

Example (Custom Config)

```
{
  "fields": {
    "full_name": "name",
    "emails": "email_addresses",
    "phones": "phone_numbers",
    "skills": "skills",
    "experience": "work_history",
    "salary_expectation": "expected_ctc"
  },
  "include_confidence": true,
  "include_provenance": true,
  "on_missing": "null"
}
```

5. Edge Cases & Scope

Handled (Graceful Degradation)	Out of Scope	Design Goals
• Missing/empty inputs (resume, notes, CSV) • Duplicate emails / phones • Invalid emails / phones • Missing skills • Missing education / experience • Malformed input files • Partial / incomplete data	• OCR for scanned PDFs • LLM / ML-based extraction • Fuzzy entity resolution • Non-English resumes • Advanced semantic matching beyond simple identifiers	• Deterministic • Explainable • Robust • Configurable • Scalable